

D472

21st Century Operations and Supply Chain

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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Items

1.

A car-parts maker with plants in three states ran lean with 2 days of inventory. A chip shortage at its single supplier idled lines for 5 weeks. The CEO, a former engineer, wants to add buffers without abandoning lean. Which approach fits best?

- A. Hold strategic buffers for critical, long-lead, single-source parts and stay lean elsewhere.
- B. Raise inventory of every part to 60 days.
- C. Add buffers only for cheap, high-volume parts.
- D. Stay at 2 days, since lean is the cheapest approach.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

A US apparel firm pays \$10.00 per unit from Asia and is quoted \$11.40 from Mexico. Asia adds \$1.20 in freight and duty and \$0.90 in carrying cost for an 8-week transit; Mexico adds \$0.40 in freight and duty and \$0.15 in carrying cost. The Mexico plant opened in 2020. Which source is cheaper per unit landed?

- A. Mexico: \$11.95 vs. \$12.10.
- B. Mexico, which will always be the better choice.
- C. Asia, by \$1.40.
- D. Asia, by \$0.60.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A retailer plans for peak season. Option 1: pre-book air capacity for a fixed \$400,000. Option 2: wait and buy spot air only if a port disruption occurs; the chance is 30%, and spot capacity would cost \$1.2 million. The logistics director prefers Option 1 because a competitor chose it. Which option has the lower expected cost?

- A. Option 1, since \$400,000 is less than \$1.2 million.
- B. Option 1: \$400,000 vs. \$840,000.
- C. Option 2: \$360,000 vs. \$400,000.
- D. Option 1, matching the competitor's choice.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4. [TRANSFER — reversed stem / different variable]

Pre-booking capacity costs \$400,000. Spot capacity would cost \$1.2 million if a disruption happens. Above what disruption probability does pre-booking have the lower expected cost?

- A.** About 66.7%
- B.** 30%
- C.** About 33.3%
- D.** 300%

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

A pharmaceutical distributor learns about supplier delays only when shipments fail to arrive. It has 14 regional warehouses and a new logo. Which investment most directly improves its response time to disruptions?

- A.** Monthly supplier scorecards.
- B.** End-to-end visibility that links supplier, carrier, and shipment tracking data with automatic alerts.
- C.** A rebrand that signals reliability.
- D.** Larger warehouses.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

A firm scores two suppliers from 1 to 5 on cost (weight 0.5), resilience (0.3), and sustainability (0.2). Supplier P scores 5, 2, 3. Supplier Q scores 3, 5, 4. Supplier P's sales rep has visited three times this year. Which scores higher?

- A.** Q: 12 vs. 10.
- B.** P: 3.8 vs. 3.7.
- C.** P, because it wins the most heavily weighted criterion.
- D.** Q: 3.8 vs. 3.7.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7. [TRANSFER — reversed stem / different variable]

Supplier Q's weighted score is 3.8. If its resilience score (weight 0.3) falls from 5 to 3, what is its new score?

- A.** 4.4
- B.** 3.2
- C.** 3.5
- D.** 1.8

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

A specialty chemical plant would need 10 weeks to qualify a new supplier if its only resin supplier failed. It holds 6 weeks of resin, and its customers' own stocks cover 2 more weeks. The plant manager runs marathons. What is the exposure?

- A.** A 2-week gap in which customer demand goes unmet.
- B.** A 4-week gap.
- C.** No exposure, because survival time exceeds recovery time.
- D.** An 8-week gap.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9. [TRANSFER — reversed stem / different variable]

A plant holds 5 weeks of a part and uses 400 units a week. Recovering from a supplier failure would take 8 weeks. How many additional units of buffer close the gap?

- A.** 3,200 units
- B.** 2,000 units
- C.** 1,200 units
- D.** 5,200 units

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

A medical-mask maker buys all its melt-blown fabric from one low-cost supplier in one region; a regional lockdown just halted that supplier for a month. The supplier has won quality awards. What is the most resilient sourcing change?

- A.** Stay single-source because of the supplier's quality awards.
- B.** Qualify a second supplier in a different region and split volume.
- C.** Move 100% of volume to one new supplier elsewhere.
- D.** Add a second supplier in the same region.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A beverage company forecast 100, 200, and 400 cases for three months; actual demand was 80, 250, and 400. Its forecasting software is two years old. What is the MAPE, with errors taken as a percent of actual?

- A.** About 9.6%
- B.** About 1.7%
- C.** 25%
- D.** 15%

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12. [TRANSFER — reversed stem / different variable]

A month's actual demand was 500 units and its absolute percent error was 12%. The forecast was above actual. What was the forecast?

- A.** 440
- B.** About 568
- C.** 512
- D.** 560

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer A Concept 1

A car-parts maker ~~with plants in three states~~ ran lean with 2 days of inventory. A chip shortage at its single supplier idled lines for 5 weeks. The CEO, ~~a former engineer~~, wants to add buffers without abandoning lean. Which approach fits best?

Target word: buffer strategy **Decides:** Risk sits in critical single-source parts.

A	Correct	Puts resilience where the risk is and keeps efficiency elsewhere.
B	Absolute language	Ties up cash in low-risk parts.
C	Inverted operation	Protects the parts that were never the problem.
D	Dropped element	Drops the cost of the five-week shutdown.

2. Answer A Concept 2

A US apparel firm pays \$10.00 per unit from Asia and is quoted \$11.40 from Mexico. Asia adds \$1.20 in freight and duty and \$0.90 in carrying cost for an 8-week transit; Mexico adds \$0.40 in freight and duty and \$0.15 in carrying cost. ~~The Mexico plant opened in 2020.~~ Which source is cheaper per unit landed?

Target word: landed cost comparison **Decides:** Asia \$12.10; Mexico \$11.95.

A	Correct	$10.00 + 1.20 + 0.90 = 12.10$; $11.40 + 0.40 + 0.15 = 11.95$.
B	Absolute language	The edge is \$0.15 and depends on today's inputs.
C	Adjacent category	Compares factory prices, not landed cost.
D	Dropped element	Drops carrying cost: 11.20 vs. 11.80.

3. Answer C Concept 3

A retailer plans for peak season. Option 1: pre-book air capacity for a fixed \$400,000. Option 2: wait and buy spot air only if a port disruption occurs; the chance is 30%, and spot capacity would cost \$1.2 million. ~~The logistics director prefers Option 1 because a competitor chose it.~~ Which option has the lower expected cost?

Target word: lower expected cost **Decides:** $0.30 \times \$1.2M = \$360,000$.

A	Dropped element	Drops the 30% probability.
B	Wrong stem number	Uses a 70% chance instead of 30%.
C	Correct	Expected spot cost is 30% of \$1.2 million.
D	Camouflage	A competitor's choice is not a cost.

4. Answer C Concept 3 · transfer

Pre-booking capacity costs \$400,000. Spot capacity would cost \$1.2 million if a disruption happens. Above what disruption probability does pre-booking have the lower expected cost?

Target word: solve for break-even probability **Decides:** $400,000 / 1,200,000$.

A	Confusion set	The chance of no disruption at break-even.
B	Wrong stem number	A common planning figure, not the break-even.
C	Correct	$400,000 / 1,200,000 = 1/3$.
D	Inverted operation	$1.2M / 0.4M$ divides the wrong way.

5. Answer B Concept 4

A pharmaceutical distributor learns about supplier delays only when shipments fail to arrive. ~~It has 14 regional warehouses and a new logo.~~ Which investment most directly improves its response time to disruptions?

Target word: capability **Decides:** Problems are detected too late.

A	Dropped element	Lagging reports lack the real-time signal.
B	Correct	Early detection gives time to respond.
C	Camouflage	Branding does not change detection.
D	Adjacent category	More space does not reveal delays sooner.

6. Answer D Concept 5

A firm scores two suppliers from 1 to 5 on cost (weight 0.5), resilience (0.3), and sustainability (0.2). Supplier P scores 5, 2, 3. Supplier Q scores 3, 5, 4. ~~Supplier P's sales rep has visited three times this year.~~ Which scores higher?

Target word: weighted totals **Decides:** $P = 2.5 + 0.6 + 0.6$; $Q = 1.5 + 1.5 + 0.8$.

A	Dropped element	Adds raw scores and drops the weights.
B	Inverted operation	Swaps the two totals.
C	Absolute language	The heaviest criterion does not decide by itself.
D	Correct	$P = 3.7$; $Q = 3.8$.

7. Answer B Concept 5 · transfer

Supplier Q's weighted score is 3.8. If its resilience score (weight 0.3) falls from 5 to 3, what is its new score?

Target word: recompute **Decides:** $\text{Change} = 0.3 \times (-2)$.

A	Inverted operation	Adds the change instead of subtracting it.
B	Correct	$3.8 - 0.6 = 3.2$.
C	Wrong stem number	Subtracts the weight (0.3) instead of weight $\times$ change.
D	Dropped element	Subtracts 2 points without the weight.

8. Answer A Concept 6

A specialty chemical plant would need 10 weeks to qualify a new supplier if its only resin supplier failed. It holds 6 weeks of resin, and its customers' own stocks cover 2 more weeks. ~~The plant manager runs marathons.~~ What is the exposure?

Target word: gap between recovery and survival **Decides:** $TTS = 6 + 2 = 8$ weeks; $TTR = 10$.

A	Correct	$10 - 8 = 2$ weeks.
B	Dropped element	Leaves out the customers' 2 weeks of stock.
C	Inverted operation	Recovery (10) exceeds survival (8).
D	Wrong stem number	That is the time to survive, not the gap.

9. Answer C Concept 6 · transfer

A plant holds 5 weeks of a part and uses 400 units a week. Recovering from a supplier failure would take 8 weeks. How many additional units of buffer close the gap?

Target word: solve for buffer **Decides:** $(8 - 5) \times 400$.

A	Dropped element	Ignores the 5 weeks already on hand.
B	Wrong stem number	The current 5-week stock.
C	Correct	3 weeks $\times$ 400.
D	Inverted operation	Adds 8 and 5 weeks instead of subtracting.

10. Answer B Concept 7

A medical-mask maker buys all its melt-blown fabric from one low-cost supplier in one region; a regional lockdown just halted that supplier for a month. ~~The supplier has won quality awards.~~ What is the most resilient sourcing change?

Target word: sourcing change **Decides:** The risk is regional and single-source.

A	Camouflage	Quality does not reduce regional risk.
B	Correct	Diversifies the shared regional risk.
C	Absolute language	Still single-source; the risk just moves.
D	Dropped element	Both would stop in the same lockdown.

11. Answer D Concept 8

A beverage company forecast 100, 200, and 400 cases for three months; actual demand was 80, 250, and 400. ~~Its forecasting software is two years old.~~ What is the MAPE, with errors taken as a percent of actual?

Target word: MAPE **Decides:** $|20|/80 = 25\%$, $|50|/250 = 20\%$, $0/400 = 0\%$.

A	Adjacent category	70 / 730 is a volume-weighted error, not MAPE.
B	Dropped element	Drops absolute values, so errors cancel.
C	Wrong stem number	The single largest monthly error.
D	Correct	$(25\% + 20\% + 0\%) / 3 = 15\%$.

12. Answer D Concept 8 · transfer

A month's actual demand was 500 units and its absolute percent error was 12%. The forecast was above actual. What was the forecast?

Target word: solve for the forecast **Decides:** Error = $0.12 \times 500 = 60$, forecast above actual.

A	Inverted operation	Places the forecast below actual.
B	Confusion set	$500 / 0.88$ bases the error on the forecast, not actual.
C	Wrong stem number	Adds 12 units instead of 12%.
D	Correct	$500 + 60 = 560$.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owled.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	10	
Wrong stem number	A figure copied from the prompt that is not the answer.	7	
Adjacent category	The right idea in the wrong bucket.	3	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	7	
Absolute language	Always, never, guarantee — where judgment applies.	4	
Confusion set	Every option is real; the test is exact role-matching.	2	
Camouflage	Answers a stem fact that was never the question.	3	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Resilience vs. efficiency	Add buffers selectively for critical, long-lead, or single-source items; stay lean where risk is low.	Choosing all-lean or all-buffer.	A lean operation hit by a disruption.
2	Landed cost for nearshoring	Compare unit price plus freight, duty, and in-transit carrying cost, not unit price alone.	Deciding on the factory quote.	A cheaper far source vs. a pricier near source with adders.
3	Expected cost of contingency options	Expected cost = probability x cost if it happens; compare against the sure cost of preparing.	Comparing a sure cost with the full worst-case cost.	A fixed prepay option vs. a pay-if-it-happens option.
4	Supply chain visibility	Real-time, end-to-end visibility across suppliers and carriers speeds detection of and response to disruptions.	Relying on lagging reports or more space.	Learning of problems only when shipments fail to arrive.
5	Weighted scoring	Multiply each score by its weight and add; the highest total wins.	Letting the heaviest criterion decide alone, or adding unweighted scores.	Criteria weights and 1-5 scores.
6	Time to recover vs. time to survive	If time to recover exceeds time to survive, the gap is unmet demand; close it with buffer or faster recovery.	Counting only own inventory, or reading the comparison backwards.	Weeks to requalify a supplier vs. weeks of stock.
7	Geographic diversification	A second source must not share the first source's risks, such as the same region.	Dual-sourcing within the same region.	A regional shutdown hits a single supplier.
8	Forecast error (MAPE)	MAPE = the average of actual – forecast / actual across periods.	Letting positive and negative errors cancel.	Forecast and actual values for several periods.

Course scope (public description): 21st Century Operations and Supply Chain explores modern issues in supply chain management. Building on the supply chain landscape introduced in prior courses, this course addresses more complex supply chain and operations issues. The course examines how disruptions transform operations and the supply chain and how to use qualitative and quantitative data to evaluate solutions.